

LIAM TAYLOR

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Education

Bachelor of Science in Electrical Engineering, Concentration in Power Systems

Jan 2026-Present

University of South Florida, Tampa, FL

Bachelor of Science in Physical Sciences

Dec 2024

University of South Florida, Tampa, FL

Relevant coursework : Material and Energy Balances, Transport phenomena, Physical Chemistry, Modern Physics

Research Experience

Undergraduate Research Assistant, TROPICS Lab, Dept. of Electrical Engineering, USF Sept 2026 – Present

- Placed on the Noninvasive Blood Glucose Measurement Validation and Miniaturization projects in Dr. Ashwin Parthasarathy's biophotonics/tissue-optics lab, under mentor Mrinmoy Turja.

Work Experience

Assembly Technician 2, Lithionics, Clearwater, FL

Jan 2025 – Jul 2026

- Performed First Article Inspections on 400A-rated UL-certified Battery Management Systems and 12V/24V/51V/102V single-module lithium-ion batteries, verifying internal data-monitoring components and physical build integrity.
- Conducted discharge testing and final-approval inspections on multi-module battery systems, maintaining in-house data records against customer specifications and using findings to drive process-checklist improvements.
- Collaborated with engineering to redesign a power-connection voltage/polarity testing tool now standard in the production process, reducing defect rates and improving technician safety during module testing.

Assistant Foreman, TE Taylor Co., Tampa, FL

Jun 2018-Aug 2024

- Direct and coordinate multiple company leaders and crew members in commercial development of ADA-compliant concrete sidewalks, ramps, and drainage valleys
- Evaluate site conditions and project plans with head foreman to optimize workflow, reducing overtime and increasing completion rates

Personal Projects

Court Commander, *Personal Project* — *Vanilla JavaScript, HTML5 Canvas, SheetJS*

- Built a browser-based volleyball coaching application handling live scoring, automatic rotation logic, dual-team lineup management, and a 6-rotation planner with stat-tracking and XLSX export via SheetJS — used as the primary in-game tool for a competitive high school program.

IEEE EXO — Project Lead, *Robotic Gait-Assist Leg System*

- Leading a cross-organizational robotic exoskeleton project spanning IEEE EXO, a biomedical engineering club, and a campus research lab, coordinating six subsystem teams against a two-semester timeline and \$5,000 budget; authored the project's governance and requirements framework to drive accountable, independent parallel workstreams.

Mime, *IEEE EXO Flagship Project* — *Gesture-Controlled Robotic Hand*

- Developed a dual-mode (MediaPipe webcam vision and flex-sensor glove) gesture control system for a 5-finger tendon-driven robotic hand on an ESP32/PCA9685 platform, with calibration tooling and a debounced real-time gesture interface.

Copper Electroplating, *Material Science / Electrochemistry Project*

- Electroplated a copper coating onto an organic specimen using a conductive graphite-painted cathode in a copper sulfate electrolytic cell with a copper-wire anode and 6V DC source, depositing over 10g of copper in 6 hours; applied Faraday's laws of electrolysis to relate deposited mass to charge and current.

Technical Skills

Technologies: C, Python (pandas, numpy, astropy), MATLAB/Simulink, HTML, Java, JavaScript, CSS

Certification : SAcHe Certificates (Process Safety Hazards, Process Safety Intro, Hazard Recognition), CPR

Software: Bootstrap, GitHub Copilot, Microsoft Office (Excel, Word, PowerPoint)